UNIVERSITY OF EDINBURGH

COLLEGE OF SCIENCE AND ENGINEERING

SCHOOL OF INFORMATICS

COMPUTER COMMUNICATIONS AND NETWORKS

Thursday 21 August 2008

14:30 to 16:30

Year 3 Courses

Convener: G Hayes
External Examiners: J Gurd, M Wooldridge

INSTRUCTIONS TO CANDIDATES

Answer any TWO questions.

All questions carry equal weight.

CALCULATORS MAY BE USED IN THIS EXAMINATION

1. (a) Give two key differences between the Internet (TCP/IP) and OSI network architectures. [*2 marks*]

(b) Distinguish between network and application architectures. [*2 marks*]

(c) A system has an n-layer protocol hierarchy. Applications generate messages of length M bytes. At each of the layers, an h-byte header is added. What fraction of the network bandwidth is filled with headers? [*2 marks*]

(d) Suppose that x bits of user data are to be transmitted over a k-hop path in a packet-switched network as a series of packets, each containing p data bits and h header bits, with $x \gg p + h$. The bit rate of the lines is b bps and the propagation delay is negligible. What value of p minimizes the total delay? [*4 marks*]

(e) What does pipelining mean in the context of reliable data transfer protocols? Describe two well-known pipelined reliable data transfer protocols. Give an example of an application layer protocol in which a pipelining-like approach is used. [*3 marks*]

(f) Datagram fragmentation and reassembly are handled by IP and are invisible to TCP. Does this mean that TCP does not have to worry about data arriving in the wrong order? [*2 marks*]

(g) BGP and RIP are Internet routing protocols that rely on TCP or UDP for their message exchanges. Since they are conceptually part of the network layer, why do they do not just use raw IP packets? [*3 marks*]

(h) Assume that TCP implements an extension that allows window sizes much larger than 64 KB. Suppose that you are using this extended TCP over a 1-Gbps link with a round-trip propagation delay of 100 ms to transfer a 10-MB file, and the TCP receive window is 1 MB. If TCP sends 1-KB packets (assuming no congestion and no lost packets):

i. How many round-trip times (RTTs) does it take until slow start opens the send window to 1 MB? [*2 marks*]

ii. How many RTTs does it take to send the file? [*2 marks*]

iii. If the time to send the file is given by the number of required RTTs multiplied by the RTT value, what is the effective throughput for the transfer? What percentage of the link bandwidth is utilized? [*3 marks*]

2. (a) List two advantages and two disadvantages for having international standards for network protocols. Name two prominent sets of standards relevant to computer networking. [*6 marks*]

(b) An object of size S needs to be transferred over a 2-Mbps long-distance link with a round-trip propagation delay of 500 ms. At what value of the transfer size S will the transmission delay start to dominate in the overall transfer delay. Assume delays other than due to propagation and transmission to be negligible. [*3 marks*]

(c) Suppose a host has a 1-MB file that is to be sent to another host over a 1-Mbps link with 10 ms one-way propagation delay. The file takes 1 second of CPU time to compress 50%, or 2 seconds to compress 60%. Ignore queueing delays.

i. Calculate the bandwidth at which the combined time taken for compression and transmission for each of the compression options equals that of the uncompressed case. [*4 marks*]

ii. Does propagation delay also get affected? Why or why not? [*2 marks*]

(d) List four key differences between IPv4 and IPv6. [*4 marks*]

(e) Suppose a large organization with 1000 routers employs a simple 2-level routing hierarchy with the routers equally distributed between 20 areas (as in OSPF) to keep the routing table size small. Assume that, unlike OSPF, every router needs to keep an entry in its routing table indicating the route to reach areas other than the one it belongs to. At what network size in terms of the total number of routers will it no longer be possible to keep the routing table size at or below 100 entries with the above mentioned 2-level hierarchy? [*4 marks*]

(f) How is the notion of "subnets" related to hierarchical routing? [*2 marks*]

3. (a) Describe the service offered by each of the following to their immediate higher layers: TCP, UDP, IP and Ethernet. [*4 marks*]

(b) Suppose your organization decides to upgrade the existing Ethernet from 10 Mbps to 100 Mbps without changing the network configuration, node locations and cabling.

i. What would you as a network designer do to maintain the same efficiency as before? [*3 marks*]

ii. Suppose that you are also tasked with improving the efficiency of the network after the upgrade. What would you recommend your users to do? Discuss your approach. [*4 marks*]

(c) Two CSMA/CD stations are each trying to transmit long (multiframe) files. After each frame is sent, they contend for the channel, using the binary exponential backoff algorithm. What is the probability that the contention ends on round k, and what is the mean number of rounds per contention period? [*6 marks*]

(d) Give two reasons why networks might use an error-correcting code instead of error detection and retransmission. [*4 marks*]

(e) When a file is transferred between two computers, two acknowledgement strategies are possible. In the first one, the file is chopped up into packets, which are individually acknowledged by the receiver, but the file transfer as a whole is not acknowledged. In the second one, the packets are not acknowledged individually, but the entire file is acknowledged when it arrives. Discuss the relative merits these two approaches. [*4 marks*]